9.8 Taylor and Maclaurin Series

Taylor series - a power series representing a function

Series Representations

A power series is a continuous function with derivatives of all orders. What about the other way round?

Assume that a function $f(x)$ is the sum of a power series

$$f(x) = \sum_{n=0}^{\infty} a_n (x - a)^n$$

$$= a_0 + a_1(x - a) + a_2(x - a)^2 + \ldots$$

with a positive radius of convergence.

By repeatedly differentiating within the interval of convergence, we can work out the terms a_n for each $n \geq 0$.

If f has a series representation, then the series must be

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \ldots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \ldots$$

Taylor and Maclaurin Series

Definition: Let f be a function with derivatives of all orders throughout some interval containing a as an interior point. Then the **Taylor series generated by f at $x = a$** is

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \ldots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \ldots.$$

The **Maclaurin series generated by f** is

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!}x^k = f(0) + f'(a)x + \frac{f''(0)}{2!}x^2 + \ldots + \frac{f^{(n)}(0)}{n!}x^n + \ldots,$$

the Taylor series generated by f at $x = 0$.

Note: We know that the power series $\sum_{n=0}^{\infty} (-1)^n x^n$ converges to $\frac{1}{1+x}$ in the interval $x \in (-1, 1)$. So the Taylor series generated by $f(x) = \frac{1}{1+x}$ at $x = 0$ is precisely $\sum_{n=0}^{\infty} (-1)^n x^n$.

Example: Find the Taylor series generated by $f(x) = \frac{1}{x}$ at $a = 2$. When does the series converge to $\frac{1}{x}$?

Taylor Polynomials

Taylor polynomials - to approximate a function.

Definition: Let f be a function with derivatives of order k for $k = 1, 2, \ldots, N$ in some interval containing a as an interior point. Then for any integer n from 0 through N, the **Taylor polynomial of order** n generated by f at $x = a$ is the polynomial

$$P_n(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f^{(k)}(a)}{k!}(x - a)^k + \ldots + \frac{f^{(n)}(a)}{n!}(x - a)^n.$$

Example: The first few Taylor polynomials $f(x) = \cos x$ at $x = 0$ approximate the cosine function near $x = 0$.

$$P_0(x) = P_1(x) = 1$$

$$P_2(x) = P_3(x) = 1 - \frac{x^2}{2!}$$

$$P_4(x) = P_5(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!}$$

Example: Expand $f(x) = e^x$ at $x = 0$.

Example: Expand $f(x) = \ln(1 + x)$ at $x = 0$.

Example: Expand $f(x) = \sin x$ at $x = 0$.

Example - A function f with converging Taylor series, but not equal to f itself.

$$f(x) = 0, \qquad x = 0$$
$$= e^{-1/x^2}, \quad x \neq 0$$

f has derivatives of all order at $x = 0$ and $f^{(n)}(0) = 0$ for all n.

Methods to generate power series

Recall that we can substitute a function $f(x)$ for x in a convergent power series. This method can be used to generate Taylor series for complicated functions.

Example: Find the Taylor series of $f(x) = \frac{1}{1+x^3}$ at $x = 0$.

Example: $f(x) = \ln(\cos x), x \in (-\frac{\pi}{2}, \frac{\pi}{2})$

Example - Indirect Expansion:

Find the Taylor series of $f(x) = (1 + x)e^{x}$ at $x = 0$.

Example:

Find the Taylor series of $f(x) = \frac{e^x}{\cos x}$ at $x = 0$.